Knowledge Management Strategy for academic organisations

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Knowledge Management (KM) is a new scientific discipline which facilitates organisations to better perform in their highly competitive environment. Universities as learning organisations in the centre of the knowledge triangle are faced with the need to better manage their knowledge resources and to facilitate their transfer to all interested stakeholders. Similar to companies, universities are developing at different speeds, and some are lagging behind in adopting KM strategic approach. The goal of this paper is to provide guidance to university staff in developing their KM strategy and integrating it in the overall organisational strategy.

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1. INTRODUCTION

The fast changes in science and technology development nowadays shorten the learning cycle of individuals and organizations. Subsequently, organisations in the private and the public sector should strategically manage their knowledge assets in order to accelerate the creation and transfer of new knowledge, as well as to facilitate the access and usage of the available resources. The company practice has shown that knowledge, when properly used and leveraged, could drive organisations to become more innovative and thus, more competitive in their highly volatile environment [2].

The adoption of Knowledge Management (KM) as a company practice was driven by large organizations and consulting companies world-wide. KM concepts, tools and methods are helping organizations to improve decision making, working processes and knowledge sharing, accelerate innovation, reduce duplication of work, and manage more effectively their knowledge assets. Information and communication technologies (ICT), on their side, provide opportunities for better organization and access to all explicit knowledge resources, however, the tacit knowledge of people is more difficult to be acquired, formalized and used [1].

Universities are conservative organizations, often not eager to make deep changes in their working processes and practice. They are in the centre of the knowledge triangle – with their role for knowledge creation, dissemination and application. The availability of organizational strategy normally determines the overall mission, priorities and goals of the university, however, rarely focuses explicitly on knowledge management processes. The knowledge of individuals and groups is essential for all activities of the university, however, normally difficult to access. Similar to private companies, universities differ in their KM adoption, which generally affects their performance, and subsequently, their overall ranking on the high competitive global educational market.

The aim of this paper is to provide guidance for preparing a KM strategy in academic institutions. It follows the Balanced Scorecard (BSC) approach of Kaplan and Norton for organisational strategy development [7], and presents the logical steps to be followed by practitioners. Normally, before starting the strategy elaboration an analysis of the organizational environment (both internal and external) should be carried out in order to provide the basis for decision making. Thus, the paper logically follows the results of the Knowledge

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Audit (KA), carried out following another set of patterns presented at workshops of EuroPLoP 2009, 2010, 2011 and 2012 [3, 4, 5, 6].

The patterns in this paper are intended to codify business practices in the area of Knowledge Management in a pattern language so that they may be better understood, communicated, applied and studied. The previous work of the authors focused on Knowledge Audit processes and tools used (Table 1), whereas this paper provides the next set of patterns linked to the elaboration of a Knowledge Management strategy.

The patterns are intended for Knowledge Management practitioners, for university managers, researchers, and students. They may be applied in the context of knowledge-intensive public or private organizations, in particular academic institutions. For small and medium enterprises some adaptations are needed according to their specificity, and especially, in the financial perspective.

Table 1 Knowledge Audit patterns

PATTERN NAME	**PATTERN PURPOSE**
KNOWLEDGE AUDIT PLAN, [EuroPLoP'2009]	Planning of Knowledge Audit scope, activities and resources
KNOWLEDGE AUDIT TEAM, [EuroPLoP'2009]	Selecting the right Knowledge Audit Team, with the desired mix of skills and knowledge
KNOWLEDGE AUDIT METHODOLOGY, [EuroPLoP'2009]	Develop methodology for successfully performing specific Knowledge Audit tasks and activities
KNOWLEDGE AUDIT QUESTIONNAIRE, [EuroPLoP'2009]	Select, compose or adapt Knowledge Audit Questionnaire according to specific company needs
KNOWLEDGE ASSETS MAPPING [EuroPLoP'2011]	Identify, locate, and assess knowledge assets, and on this base set priorities and identify action needs
KNOWLEDGE LANDSCAPE MAPPING [EuroPLoP'2011]	Assess available KM practices, programs, projects, infrastructure elements, policies and procedures, etc., and determine actions for improvement
KNOWLEDGE FLOWCHARTS [EuroPLoP'2012]	Identify existing paths, means of knowledge flows between individuals, groups and in the organization as a whole, aimed at improving knowledge flows
COMPETITIVE KNOWLEDGE ANALYSIS [EuroPLoP'2011]	Identify areas of expertise and important knowledge assets providing competitors strengths and opportunities
KNOWLEDGE DIAGNOSTICS [EuroPLoP'2012]	Understand knowledge-related mechanisms and processes, both at individual, group and organizational levels
CRITICAL KNOWLEDGE FUNCTION ANALYSIS [EuroPLoP'2012]	Identify critical operational, professional or managerial functions, and determine the potential value of their knowledge-related improvements
KNOWLEDGE MANAGEMENT BENEFITS ASSESSMENT [EuroPLoP'2011]	Focus on potential effects of KM initiatives as a base for planning, action, and monitoring of KM implementation
KNOWLEDGE AUDIT DATA GATHERING [EuroPLoP'2010]	Ensure high-quality data gathering and involvement of staff in the KA process
KNOWLEDGE AUDIT ANALYSES OF RESULTS [EuroPLoP'2010]	Analyse the Knowledge Audit results, test and verify hypothesis based on the collected quantitative and qualitative data
KNOWLEDGE AUDIT REPORTING [EuroPLoP'2009]	Prepare a meaningful Knowledge Audit Report as a major outcome of the Knowledge Audit process
STRATEGIC ROADMAP (EuroPLoP 2014)	How to implement Knowledge Management – a set of patterns to be developed on KM strategy

2. THE PATTERNS

The KM strategy development is based on the results of a Knowledge Audit carried out in the organization and involving its staff in assessment of the knowledge state-of-the-art at individual, group and organisational levels. The outcomes of the KA provide the necessary information for all processes in the strategy development, and each process is linked to a pattern to be followed by practitioners (Figure 1). This paper proposes the following patterns each linked to a step in the strategy development process:

- KNOWLEDGE MANAGEMENT MISSION – Defining the KM purpose in order to ensure the organisational mission fulfilment (why you need KM);
- KNOWLEDGE MANAGEMENT GOALS – Defining KM goals in order to achieve the KM mission (what KM should achieve);

- Knowledge Management Strategic Choices– Choosing the most suitable for the organization strategic alternative (how to implement KM).

Generally, the mission clarifies the reasons for strategy development, and provides a sound base for determining the strategic goals and priorities. Once this is clarified, it should be decided how to implement the goals and which actions would be suitable. The next logical step is to prepare a plan linking goals with the consequent activities, resources and timeframe. At the same time, a crucial issue is the monitoring and control of the implementation, measuring the progress with predefined key performance indicators (KPI). The last steps of the strategy development process will be considered in a separate paper.

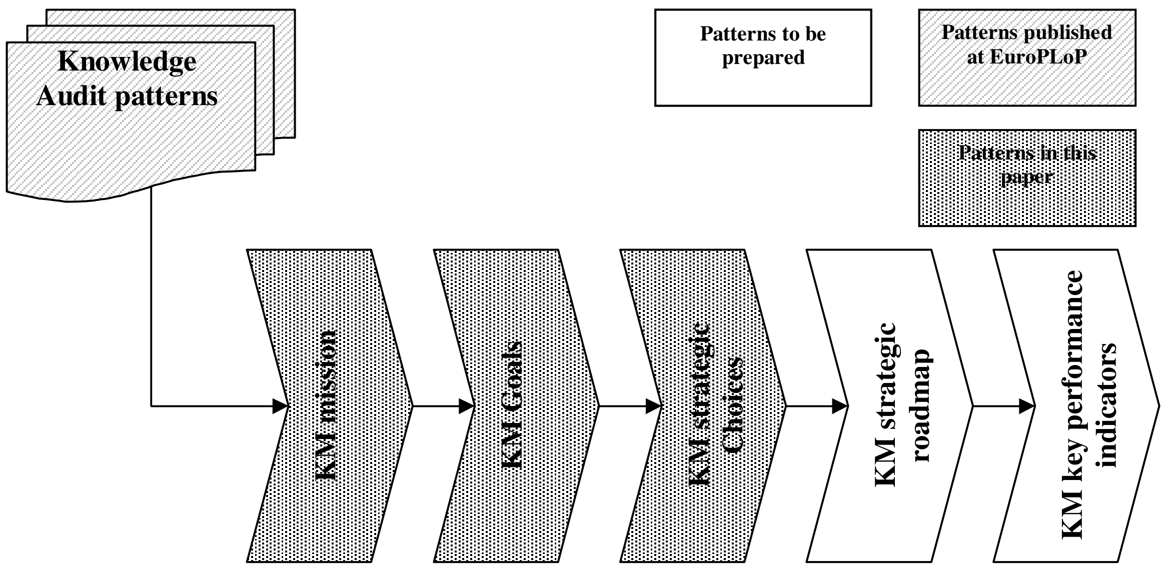

Figure 1 Knowledge Management strategy processes

3. KNOWLEDGE MANAGAMENT MISSION

In its strategy a university set its mission and vision, and determined its strategic goals and related actions. University performance highly depends on the management of its knowledge assets. The awareness on the knowledge resources and the research results of its staff is essential for better ranking and higher assessment of each university. At the same time, the new knowledge created by researchers and university teams could be used for practical exploitation and gaining financial benefits from valuable intellectual assets. Making a comprehensive Knowledge Audit is a necessary step for evaluation of the knowledge state-of-the-art. Next step is to decide how to strategically manage knowledge assets.

3.1. Problem: Which should be the purpose of KM in the organisation?

Knowledge management could help organizations to implement their strategic mission and vision, and improve their overall performance. The KM mission generally sets the reasons (purpose) for undertaking KM, and it is linked to the vision 'how organizational KM should be in the future'.

- *KM scope:* KM could include comprehensive measures covering the whole organization including all working processes, all units and levels (strategic, managerial or operational). Such KM strategy requires heavy investments, often not linked to better performance and meeting needs of employees.
- *Internal or external focus:* KM could support the organizational performance by focusing on the needs of its customers (external stakeholders) and providing them better services. Another option is the development of different competences then those of competitors.
- *Type of knowledge:* There are a large variety of knowledge resources, some of which is not worth managing. At the same time, depending on the type of knowledge (collective or individual, tacit or explicit) different tools and approaches could be applied. A comprehensive KM approach might be too expensive, not necessary and not sufficiently effective.

3.2. Solution: Set the mission according to the knowledge needs of the organisational strategy

Analyze first the overall organisational strategy and the knowledge-related problems (gaps) identified in the KNOWLEDGE AUDIT REPORT [3]. Consider that KM should help the organization achieving its mission and strategy. For this purpose:

- Take into account the results of the KNOWLEDGE ASSETS MAPPING [5] and the KNOWLEDGE DIAGNOSTICS [6] in order to determine on which level of the organisation to focus and on which type of knowledge.
- Analyze the results of the CRITICAL KNOWLEDGE FUNCTION ANALYSIS [6], in order to determine the working processes and organisational functions which should be supported.
- Consider the organisational strategy and the COMPETITIVE KNOWLEDGE ANALYSIS [5] in order to decide where to invest - to strengthen your unique competences and organisational knowledge (specific learning and growth measures) or to devote more efforts for managing knowledge about customers (better understanding customers' needs).

Before taking a final decision on KM mission consider the results of the KM BENEFITS ASSESSMENT [5], and evaluate the expected KM investments and its expected impact on the organisational knowledge state.
Formulate the KM mission as a short statement to be clearly understood by the staff.
Describe the expected future KM state in the organization (the KM vision) as a base for awareness raising among employees during the following implementation phase.

3.3. Consequences

Defining the KM mission is a first step for strategy elaboration, and helps creating a vision for the expected future state of KM in the organization, and raising awareness of employees, as well as convincing the top management for the expected KM benefits.
Targeting the KM strategy at specific knowledge resources needed in the working processes and the tasks of the staff will help, on the one side, to involve and motivate employees, and as a consequence, to ensure better performance of the organisation as a whole.
The KM vision describing the future picture will help motivating the staff for implementing the KM actions.
It would be difficult to choose the most appropriate tools and methods for KM, if the type of knowledge resources on which to focus is not clearly determined.
Without a clear KM purpose it might be difficult to determine concrete KM goals and actions, and to ensure the expected KM benefits.

3.4. Example

A team at the Faculty of Mathematics and Informatics (FMI) of Sofia University (SU) prepared a strategic framework for FMI future development and outlined where to make strategic investments. Initially, some strategic questions were taken into account [10]:

- Where is the university now, not where do we hope that it is?
- If no changes are made, where will the university be in three years time?
- If the answers of first two questions are unacceptable, what specific actions should be taken?
- What risks and payoffs are involved?

The combination of various management, foresight and creativity tools and techniques (SWOT analysis, environmental scanning, driving forces analysis, scenario planning, focus group, brainstorming, etc.) helped the team for strategic thinking (exploring options), strategic decision making (making choices) and strategic planning (taking actions).
The FMI team followed the best practice of university leaders. At the same time, it was taken into account that FMI is a knowledge-intensive organisation, and its overall mission as an academic institution depends on proper managing its knowledge processes. Thus, the KM strategy coincides with the overall FMI strategy. The FMI Research, Technology Development and Innovation strategy was defined following the BSC methodology [7]. More specifically, the team determined first the mission and vision for FMI as follows [10]:

- FMI Mission: Provision of a flexible, multi-disciplinary environment for high quality informatics research and education using latest technologies and world expertise, and achieving excellence in research and teaching within the university, the country and the international ICT communities.
- FMI Vision: Creating a Faculty community that becomes locally and internationally recognized in research, teaching and service, and recognized nationally for student excellence.

4. KNOWLEDGE MANAGEMENT GOALS

The organisational strategy determines the long-term goals for development of the organization. According to the BSC approach of Kaplan and Norton [7], goals could be set in four main perspectives – financial, customers, internal processes, and learning and growth. Strategy implementation depends on overcoming the strategic gaps in each perspective, whereas the KM strategy supports overcoming the related knowledge gaps and ensuring appropriate knowledge supply to the working processes. As a knowledge-intensive organisation a university faces the need to implement knowledge management for ensuring better performance. It made a comprehensive Knowledge Audit, and decided which should be the KM mission for the organization. Having clear scope of KM, the next step is to determine the goals of the KM strategy.

4.1. Problem: What KM should achieve to facilitate the organizational strategy?

Following the BSC approach [7], and aligning the KM strategy with the overall organisational strategy, the following perspectives might be considered:

- *Financial perspective:* The more efficient use of the available knowledge resources and avoiding duplication of work by knowledge re-use could help improving organisational performance. At the same time, KM requires additional investments which now always add value to the organisation.
- *Customers' perspective:* KM could help improving services for customers by better knowing their needs. On the other side, the customers' knowledge could be used for joint value creation in an open innovation process.
- *Internal processes perspective:* Knowledge processes could facilitate the working processes if the appropriate 'hard' (ICT) or 'soft' (group facilitation and creativity techniques) tools are chosen. The choice of tools could vary according to type of knowledge processes in place (e.g. knowledge capturing, knowledge storage, knowledge sharing or knowledge re-use), and the type of knowledge resources to be managed (tacit or explicit knowledge).
- *Learning and growth perspective:* Organisations learn from external and internal sources, however, the knowledge supply highly depends on people attitude to share their knowledge and expertise. Enhancing the organisational memory might need not only better organisation of the available knowledge resources, but also targeted efforts for individuals learning and improving competences.
- *Cohesion of all perspectives:* The BSC perspectives are closely related and setting goals independently might be contra-productive.

4.2. Solution: Critically assess the knowledge gaps and determine which initiatives will help ensuring the KM mission

Use the Balanced Scorecards approach of Kaplan and Norton [7] as a framework to set KM strategy goals aligned with the KNOWLEDGE MANAGEMENT MISSION.

For the financial perspective, take into account the organisational strategy choice and the KM BENEFIT ASSESSMENT [5] results. On this base decide how KM will really support the organization, its working processes and employees and will bring the expected benefits.

Consider the KNOWLEDGE MANAGEMENT MISSION in order to set goals within the defined scope for the KM project to support the working processes and the organizational levels chosen, as well as to ensure the type of knowledge needed in them.

Take into account the organisational strategy, the results of KNOWLEDGE ASSETS MAPPING [5] and the financial perspective goals for determining the KM customers-related goals (providing them greater choice with high quality, better functionality and services). Identify the existing knowledge about customers and coming from them, and the further knowledge needs to meet the organisational strategy goals.

The KM goals related to internal processes should facilitate knowledge flows according to the respective needs of the working processes of the organisation. Take into account the knowledge-related gaps of internal environment identified in the KNOWLEDGE AUDIT REPORT [3], and on this base decide what kind of changes should be made. More specifically:

- On bases of KNOWLEDGE DIAGNOSTICS [6] decide which are the knowledge needs and the knowledge processes to be strengthened in order to support the critical business functions of the organization.
- On bases of KNOWLEDGE FLOWCHARTS [6] decide how to use existing knowledge flows in order employees to be provided timely with the knowledge they need.

- On bases of Knowledge Landscape Mapping [5] consider what should be changed in the internal practice, procedures and organization in order to facilitate KM, and to ensure employees commitment to the related KM changes.

Focus the learning and growth perspective on facilitating working processes and customers relations. Take into account the results of Competitive Knowledge Analysis [5] and Knowledge Assets Mapping [5]. On this base set learning goals for ensuring the necessary knowledge for being a capable competitor, as well as better meeting customers demands.

Do not forget to check the cohesion and complementarity of the goals set:

- Learning and growth goals should support the business processes and customers relations with the necessary knowledge and competencies.
- Internal processes goals should ensure more efficient knowledge processes, better customers relations and innovation, as well as facilitate internal social processes.
- Customers related goals based on proper internal processes, knowledge and competencies, should reflect on the financial results

4.3. Consequences

By ensuring cohesion and complementarity of the goals set for each of the four perspectives of the BSC, the following consequences are expected:

- Learning and growth goals will strengthen the knowledge and working processes and the customers relations with the necessary knowledge and competencies.
- Internal processes goals will ensure more efficient working processes and innovation processes, better customers understanding, as well as will facilitate the internal learning, social and organizational processes.
- Customers related goals based on proper internal processes, knowledge and competencies, will bring higher financial results (higher productivity and growth) by better using the available knowledge assets and by providing added-value for the clients.
- Linking the KM goals to the knowledge gaps identified during the KA, will imply on better targeting the KM strategy and the related actions.
- Having concrete goals will facilitate the whole KM process, and in particular to determine the necessary KM actions and the key performance indicators for their monitoring.
- Failing to properly consider all perspectives and the correlation between them could have a consequence in missing the expected KM benefits.

4.4. Example

The strategy modeling process at the English Language Faculty of Engineering (ELFE) of the Technical University – Sofia (TU) used business process management tools and created a BSC system as a value-added chain process (Figure 2). The most important reason for using the BSC approach was that it easily links the objectives set with related measures, targets and key performance indicators for monitoring of the implementation results.

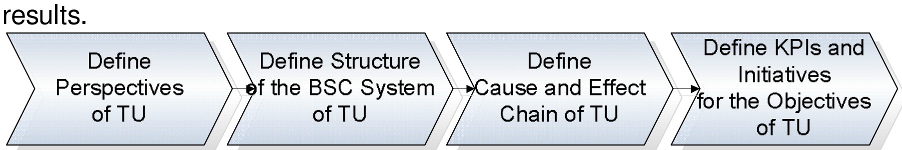

Figure 2. BSC Value-added chain diagram [10]

For the purposes of the KM strategy of ELFE was developed a strategy map consisting of three perspectives, and a specific number of strategic goals, that were related to each other in a hierarchy. The main goal was, thus, broken down into sub-goals. As ELFE is a knowledge-based publicly funded organization, the financial perspective was not elaborated. However, it was considered that the improvements in all other perspectives will reflect on higher inflow of students, possible new collaborative projects which will bring additional benefits, including more financial resources.

The *internal processes perspective* focused on improving the research environment, facilitating knowledge gathering, sharing and reuse, intellectual property (IP) and quality management:

- Improving research and educational environment
- Improving knowledge processes

- Improving quality of research and education
- Improving management of intellectual property (IP), and knowledge transfer

The *stakeholders perspective* focused on collaboration and knowledge exchange, and attractiveness of ELFE for young people:

- Enhancing international networking in research and education
- Enhancing networking with industry
- Increasing the ELFE visibility
- Improving the image of researchers in the society

The *learning and growth perspective* focused on enhancing knowledge and capabilities at individual (by learning) and organizational (through innovation) levels, and included the following goals:

- Developing a vision for future research
- Strengthening human resources
- Enhancing knowledge in the research field
- Enhancing capacity for IP protection, innovation and organisational management

5. KNOWLEDGE MANAGEMENT STRATEGIC CHOICES

The overall organisational strategy set priorities and goals for long-term development. The organization determined the mission and the goals of its KM strategy. The next step is to decide how to achieve these goals, to set priorities and to choose among possible implementation alternatives.

5.1. Problem: How to achieve KM goals?

The Knowledge Management strategy is linked to the overall organisational strategy and supports its implementation by filling in the knowledge gaps in order to reach the goals of the organization. There are various alternatives to implement KM and meet the goals set:

- *Knowledge codification or personalisation.* Some of the dilemmas of organizations are linked to the decision how to better support knowledge creation and sharing needs within different organizational groups. They could concentrate their efforts on a codification strategy (e.g. storing codified knowledge in databases or other repositories) or a personalization strategy (e.g. based on knowledge sharing among employees and better using their personal networks).The practical implementation [8], [9] shows that the mix of KM strategies (codification and personalisation) gives worse results then the focus on one of them.
- *Bottom-up or top-down approach*. Teams and individuals often enjoy the autonomy of choosing technologies, methodologies and building their own knowledge repositories (typical for academic organisations). However, other staff members and the organisation as a whole could not take advantage of these resources. Developing organisational knowledge systems is more expensive and could not meet the needs of employees, as well as individuals are not always willing to contribute and share their own knowledge.

5.2. Solution: Determine the KM methodology on bases of critical assessment of the organizational specificity, its practice and knowledge resources.

Do not forget that each organization has its specificity, and a specific KM strategy could bring high benefits in one organization, but might not work in another one. The decision should be made according to the characteristics of the organization, its products, the nature of the employees' problem solving activities, as well as the needs for knowledge reuse. Therefore:

- Consider first your organizational environment by using the results of KNOWLEDGE LANDSCAPE MAPPING AND KNOWLEDGE DIAGNOSTICS.
- Take into account the CRITICAL KNOWLEDGE FUNCTION ANALYSIS, the type of working activities and the KNOWLEDGE MANAGEMENT GOALS.

These two steps will help you to decide if you need to invest into technology infrastructure and knowledge codification (if regular knowledge re-use is in place), or to facilitate knowledge sharing among employees (personalization strategy) with specific measures for changing organizational culture and human resources policy.

Take into account that organizations aimed at innovation and developing unique or customized products have better results using the personalization strategy, while organizations dealing with similar problems, and developing more standardized mature products build their success on codification strategy [8].

List KM goals and prioritise them according to the expected benefits for the organisational strategy and the available resources.

Consider the specific internal environment and align organisational KM priorities with those of individuals and groups. Take into account also which are the most beneficial KM processes for individuals and their performance, and start with them.

5.3. Consequences

The informed strategic choice made will ensure a better KM strategy implementation results and higher benefits at all organisational levels.

The intensive use of technologies for knowledge codification (explicit KM) and knowledge reuse helps building organizational knowledge bases and ensuring technology support for working processes.

The focus on tacit KM (personalisation strategy) and knowledge sharing requires fewer investments in technology and utilisation of creativity and group management techniques, and could be more cost efficient.

Developing a comprehensive KM strategy with heavy technology investments often does not bring the expected benefits, especially if the commitment of employees is not ensured and technologies are not linked with the working processes in place.

In academic environment, not aligning organisational goals and priorities with those of individuals could lead to severe problems in the implementation, and strong staff resistance to KM changes, which due to the academic freedom could be hardly resolved.

5.4. Example

The strategic choice made by ELFE (see 4.4. above) clearly shows that in university environment a personalization strategy is more appropriate. For a knowledge-based organization like TU it is quite natural to choose a proactive strategy and develop new knowledge inside. At the same time, the technology should support all internal processes and external communications. In ELFE a special emphasis is given on:

- personal knowledge management – with measures for training, mentoring and supervision, staff exchange, access to research databases and scientific infrastructure;
- facilitating knowledge sharing – via various internal and external events (seminars, workshops, conferences, brokerage events) and using technology support;
- enhancing networking with external stakeholders – industry, other research organizations and universities, as well as potential students.

6. CONCLUSION

The paper presents three different patterns corresponding to different processes for developing a KM strategy. This ensures that the Knowledge Management strategy and the related roadmap will be better focused and will facilitate organizational performance and competitiveness. The patterns can be applied in knowledge-intensive organizations as they propose guidelines for solving specific problems of linking knowledge and working processes, better usage and provision of knowledge when and where it is needed and taking advantage of existing knowledge flows. The examples provide a bases for academic institutions to develop their own KM strategy.

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